IV/PLAY USER GUIDE

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OVERVIEW

IV/Play (pronounced 'Four Play') is a desktop/keyboard-oriented GUI front-end for MAME™. It was designed and its coding commissioned in 2011 by original MAMEUI team member John IV, from his 2006 concepts. It has a particular feature-set, is keyboard driven, and utilises many of the navigation and keyboard shortcuts of MAMEUI. It targets native Microsoft Windows, with Windows 11 preferred. IV/Play leaves emulator configuration to MAME's own ini files and passes launch arguments to the linked executable. HBMAME, MAMEUI and differently named MAME-compatible builds can also be linked; the linked build is identified in the title bar and **F1** About. IV/Play is self-contained and portable for easy transport and mobility.

In 2025 a modernization was undertaken of the 14-year-old codebase to update from its .NET 4.5 WPFWrapper origins to its current state, rebuilt entirely using WinForms, with WPF fully deprecated. It provides a high performance, RAM based experience on i7-12700K, 64GB RAM, GPU level machines. The goal is near instant launch, game list population, and zero slowdown displaying any art assets (icons, snaps, flyers, bkgrounds, etc.) utilizing as much RAM and graphic (GPU) hardware as needed; employing modern design techniques and technologies.

Naming conventions: The application name is 'IV/Play' but due to WinOS file-naming limitations, 'IV-Play' is used where necessary.

In the 1080p 96DPI era, the initial IV/Play dimensions were 1015x432; a 2.35:1 aspect ratio in honor of the 1960's TohoScope tech used in the filming of Godzilla movies. Today, on 4K monitors, the same ratio is used but at a default of 1522x648 (144dpi and 1.5X scaling).

IV/PLAY VS. MAMEUI

MAMEUI is an integrated UI to MAME and is compiled using the base source. It was the first 'official' MAME port to the Windows platform and the most popular version of MAME for decades. It leveraged the underlying MAME code and depended on it to produce its UI and functionality. This was its strength and weakness. As MAMEUI matured (b. 1997), the original development team waned over time. It often took long stretches to catch up with the sweeping core changes that periodically took place in the underlying MAME engine. While the number of supported arcade games, gambling 'fruit machines', computers, chess games, calculators, consoles, and miscellaneous other ROM based electronics neared 40,000 sets, the UI slowed down when all the icons and screenshots were processed by the near 3-decade old controls. This introduced memory leaks over time and the user experience degraded. Changes to the core at the time also resulted in huge memory footprints and 5x longer load times.

IV/Play was born out of the desire to continue a similar 'classic' desktop interface to MAMEUI without the attendant issues of being coupled to the core.

REQUIREMENTS

Windows 11 is the preferred platform. Faster modern processors and solid-state drives (SSDs / NVMes) help considerably with image/resource loading speed. IV/Play is DPI aware and scales its UI and elements for high-resolution monitors. It ships as a self-contained Native AOT (Ahead-of-Time) .NET 10 executable, so no separate .NET runtime download is required. Its GPU-based rendering and caches are designed for modern systems with ample RAM and a capable GPU.

INSTALLATION

The most straightforward installation option is to place the IV/Play files into the top level of an existing MAME directory structure alongside the respective .exe. IV/Play will utilise the structure's icons, art types, and bkground folders of an old MAMEUI installation. IV/Play does not rely on registry settings or an install so it is easily portable on external media like a thumb drive with MAME and attendant files. IV/Play will create its owned named directory under its location for its various assets, but also maintain a same level favorites.ini file, custom game list and its *.cfg file, which allow them to survive the asset directory removal during testing or refresh. The attendant e_sqlite3.dll must also be present with the .exe.

RESOURCES

The current build, its recommended icons, and snapshots for IV/Play are found at the IV/Play homepage.

MAJOR APPLICATION FUNCTIONS

IV/Play is, at its core, a game launcher. It imports XML from the linked MAME-compatible executable to populate its lists and properties, then passes back the selected game's launch command. Configure the emulator in its own ini files. HBMAME uses hbmame.ini and MAMEUI uses mame.ini. For other builds, IV/Play uses an executable-named ini when present beside the executable, otherwise mame.ini. The **F1** command-line override can supersede those settings for ordinary game launches.

IMPORT XML DATA FROM MAME.EXE

IV/Play resides in the same directory as the MAME executable or in its own directory. Upon initial launch, it searches the local directory for a MAME version. Failing to find one a dialogue box appears asking to locate the MAME.exe to 'link'. It then grabs the output of the (currently) 324MB MAME XML file (MAME.exe –listxml). **F5** refreshes the XML using the same named MAME.exe, (when a new version of MAME is released for example) while **F4** re-opens the dialogue box to select a different MAME.exe. If an **F4** import fails, IV/Play restores the previous MAME link and setup while keeping the displayed results. Copying a folder that contains both IV/Play and MAME uses the copy's MAME even while the original still exists. Links to MAME elsewhere stay external; separate installations may share an asset folder.

CONFIGURATION SCREEN (**F1**)

F1 brings up the configuration screen. Settings are stored in the user-editable IV-Play.cfg file, viewable with **F3**. **Apply** adopts the current choices without closing; **Cancel** returns to the opening settings or the last **Apply**, including View, Sort, Icon Scaling, colours and the session theme. Newly chosen art, icons and DAT sources take effect on **Apply** or **OK**. Settings scroll when needed and the footer stays available, with font and colour pickers fitting the supported 1280×720 effective desktop. **F1** keeps the highlighted game if it remains in the resulting list and waits for existing icon work before opening. Pressing its configured shortcut again closes it as **Cancel** does.

DISPLAY XML DATA IN UI

TITLE BAR

The title bar identifies IV/Play and the linked emulator build and reports the rows currently displayed. HBMAME and MAMEUI retain their recognized names; other builds use the executable name without .exe, in uppercase, so megamame.exe appears as MEGAMAME. The names mame32.exe and mame64.exe display as MAME. The name, version and build date come from the last successfully imported build; a failed import keeps the previous identity. Machine, software and favorite counts change when the new results appear, with thousands separators; an unsuccessful refresh keeps the previous displayed counts.

On Windows 11, the title bar and outer window border follow IV/Play's current theme by default while retaining the native Windows frame and controls. Title text automatically uses black or white, whichever has better contrast with the title-bar background. **F1**'s Use Themed Window Borders toggle enables or disables this behaviour; the cfg-only colour-source option below selects the theme colour used.

GAME LIST

The game list displays all the games from the MAME XML output. Clones are automatically offset from their parents (but can have the indent removed in the **F1** screen). The font is selectable as is the colour of the parents, clones, non-working, and favorites. Navigation through the list is via regular direction keys like **up**, **down**, **page up page down, home and end**.

Display of the year, manufacturer, and machine (internal MAME) name is configurable through **F1**.

Free typing 'donkey kong' in the game list goes to the first instance depending where focus is when typing; if there are no instances downward it will go to the first instance of 'donkey kong' which may wrap around to the top in the favorites.

Alt-P or **TAB** cycles through Regular List, Large List, Flat View, Grid with art, and Grid without art. **Shift-TAB** cycles backwards. If the cycle starts with art set to None, it skips the grid-with-art step. Flat View supports global sort; see the 2.6 feature section.

In a list view, hover briefly over the selected row's clipped description to see its complete text. The game list must have focus; navigation or moving away dismisses the tooltip. This does not apply to Grid or recording-history views.

The Available gamelist, selected in **F1** or via **Alt-Insert/Delete**, lists the games whose files are on the linked emulator's rompath. **Shift-F12** builds the initial audit, after which live monitoring keeps it current as archives come and go; press **Shift-F12** again after adding or removing CHDs, or if Available asks for it because the saved audit is damaged. IV/Play reads hbmame.ini for HBMAME and mame.ini for MAMEUI; other builds use an executable-named ini beside the executable, otherwise mame.ini. It follows the emulator's inipath as MAME does, and without a configured rompath uses the emulator's default roms directory. Detection reads names only, never file contents. A game is listed when its archive is present. In a merged collection a clone is listed from its parent's archive and a game also needs its BIOS archive; when IV/Play cannot tell how a collection is organized it lists the game anyway, so a game can occasionally appear that still lacks a file. A game that needs a CHD is listed only when a folder named after it or its parent, holding a .chd file, is on the rompath. Media that MAME marks as never dumped or optional is not required, and machines that need no file at all, such as Pong, are omitted.

Mouse Back retreats through the current recording, overlay, family, software-list or search layer. Mouse Forward enters the selected family or software list, or reopens the same game's recording history after Back. These buttons do not launch games. Back stops a running benchmark only over its window; over the game list it takes the list's own step.

FAVORITES

IV/Play supports the use of a .\favorites.ini file. This is a text file with one game per line, for example:

10yard
1941
dkong
frogger

Favorites are toggled by **Alt-D** and can be set in **F1**. They appear above the MAME game list in alphabetical order, without clone icon indentation, and can use separate colours. With at least one favorite, **Alt-D** cycles through Games Only, Favorites & Games, and Favorites Only. With no favorites, Games Only is the sole available mode; **Ctrl-D** adds the selected game. Removing the last favorite, including by editing favorites.ini externally, returns the view to Games Only. A filter with no matching favorites can still produce an empty Favorites Only result when the collection itself is non-empty. Non-working and clone favorites survive the visibility filters applied to the lower game list.

Favorites are added by **Ctrl-D** from the game list and removed by **Ctrl-D** from the favorites area.

Up/**Down**/**Left**/**Right**/**Page Up**/**Page Down** behave the same way they do in the game list. A single press of **down arrow** on the last favorite goes to the first regular game list item. **Home** and **End** buttons on the keyboard provide a two-stop navigation system to move between the favorites and main game lists efficiently.

- **End Key**: When the focus is in the favorites list, the first press of **End** jumps to the last favorite. A second press jumps to the very end of the main game list. If the focus is already in the main list, it jumps directly to the end.

- **Home Key**: When the focus is in the main game list, the first press of **Home** jumps to the top of the main list (directly below favorites). A second press jumps to the top of the favorites list. If the focus is already in the favorites, it jumps directly to the top.

ICONS

Each game supported by MAME can have an associated, freestanding *.ico in the .\icons directory. IV/Play also supports having all the icons in \icons\icons.zip or \icons\icons.7z.

Base icon size and icon scale presets are chosen in **F1**. Adding, changing or removing an Icons folder or archive takes effect on **Apply** or **OK**. Existing icons stay visible until a usable replacement is ready; a failed rebuild shows a warning and keeps them when publication has not occurred. Software can use icons named for its own software list, even when that list is named differently from the system that runs it.

If 'Draw non-working overlay' is toggled on in the configuration menu, a small red circle 'x' will be overlaid on any non-working game in the lower right-hand corner of its icon.

ART AREA

The upper right of the game list is the art area. The art area displays the current art type for the selected game. These are freestanding *.png files that reside in their respective directories and take the name of each game, e.g. dkong.png. IV/Play also support archived art assets so \snap\snap.zip or \snap\snap.7z are viable options.

Artwork and icon ZIP or non-solid 7z archives can be replaced by an archive-maintenance tool while IV/Play remains open. The open archive and cached pictures keep their previous contents until **F5** reloads the sources and rebuilds the icons. A tool that tries to overwrite an open archive in place may still be blocked; replacement-based updates are supported.

Alt-1 through **Alt-0** selects an art type; None is **Alt-0**. Add picture-art folders or archives in **F1**. If no picture art was available, adding the first one selects it automatically (Snap preferred); a None deliberately selected while picture art is available is kept. Sorting or cleaning the art list retains the selected folder. Removing the selected art falls back to Snap, then another picture-art entry, then None.

Opacity for the art types and art border is set in **F1** configuration dialogue.

The border width and colour for the art can be set with **F1**.

Left clicking on the art area with the cursor toggles the art type.

If the selected art source has no image for a game, IV/Play tries another configured Snap source. If none supplies an image, the background shows through. Art type None stays empty.

IV/Play supports History.XML and MAMEInfo.DAT support files available at their respective sites. They are searched for in the MAME.exe directory structure on initial IV/Play launch and can be added at any time through the **F1** configuration menu. These files' information is accessible with the 'Peek' option using the **tilde** key (**~**) above the **TAB** on most keyboards. **Left** and **right** in DAT peek mode sizes the peek overlay horizontally, the **TAB** key cycles through the DAT types while the peek overlay is open.

Snapshots and other art assets are placed in the upper right of the window offset from the edge, and are scaled to the height of the window. If art is wider than 40% of the horizontal width of the window, it will be scaled to fit. Border width, colour, and its opacity are set in the **F1** screen.

TEXT FILTER DIALOGUE

The Text Filter dialogue box is accessible with either **Ctrl-F** or **Ctrl-E**, and is extremely flexible. When typing in this field and hitting **enter** it will search and display resulting hits, like a filter. For example, typing 'donk' would bring up Donkey Kong and all other items with 'donk' in its description including something like 'Crazy Donkey'.

Search by manufacturer, year, driver source and other XML or DAT fields, or combine ordinary words with the advanced syntax in Appendix 1. Results stay until the filter is cleared. **Escape** closes the dialog; **Escape** on the game list backs out of the filtered view. The empty dialog opens a Quick Start preview with 12 examples and the latest search. **Alt-Down** or the hint line opens the full example library; More recent searches… opens up to 20 distinct saved queries in the same scrollable popup, including valid searches with no matches. The popup uses the roomier side above or below the text box and stays inside the usable desktop. **TAB** accepts an available suggestion, otherwise reaches the buttons; **Shift-TAB** moves backward. **Home**/**End** move the text caret. Completion resumes

for the next term after field:value; finish quoted values first. A malformed query shows one error and keeps the existing results, and an unknown field is named. Classic Filter in **F1** restores the classic discovery layout. The dialog retains its local **Escape** and **Alt-Left** dismissal keys.

BACKGROUND SELECTION AND RANDOMIZATION

The bottom art layer of the main list area takes a user defined background *.png image from the .\bkground\ directory. The **F1** configuration dialogue toggles the background image rotation, so IV/Play will use a random *.png from the .\bkground directory on each launch. **Note**: If the bkground image is 'larger' than current game window it will scale to fit the inner window. If the bkground image is smaller than current game window and less than 903px wide and 800 tall, it will tile from upper left. These bounds can be altered in the *.cfg.

PROPERTIES VIEW

Alt-ENTER displays the properties dialogue for the selected game. It can be dismissed with the **ESC** key. The text in each field is highlight-able and can be copied. For software, the dialogue shows the item's own details and its support in MAME (Supported, Partially supported or Not supported), followed by host rows naming the system it runs on, that system's status, source file and display.

LAUNCH GAME

Enter or a double-click launches the selected game through the linked MAME executable; MAME owns emulator configuration. Normal launches record an input session only when Auto-Record Input (.inp) is enabled in **F1**, off by default. Each recording gets a unique dated AM/PM filename, keeping earlier sessions. **Ctrl-Enter** works even when recording is off: one or more completed sessions open a recording list, newest first; none produces a notice and launches nothing. Software recordings distinguish the software list and item. Hold **Shift** while pressing **Enter** on a software-list machine to launch the system itself without media.

In the recording list, **Up/Down**, **Home/End**, **Page Up/Down**, the wheel or a click select a session; **Enter** or a row double-click plays it. **Backspace**, **Escape** or right-click returns to the same game view. Rows show aligned date/time and filename/folder columns, with the game's icons and artwork retained; the title bar shows the game and recording count. **Ctrl-Delete** permanently deletes the selected recording file and refreshes the list automatically. Deleting the last session returns to the game; a failed deletion leaves the row in place and reports the failure. Existing recordings are not renamed or automatically pruned, and a session still being recorded by IV/Play is not offered for playback.

ACCELERATOR / NAVIGATION KEYS

The following are the default keyboard shortcuts. Editable bindings may be changed in IV-Play.cfg; **F12** shows the grouped reference for the bindings currently in force. Notices that name a configurable shortcut show its current binding. Overlay find controls and the filter dialog retain their own local editing and dismissal keys.

~ (Tilde)	Toggles the DAT peek feature. Use with **Ctrl+left** and **right arrows** to size from 40 to 75 and 100 percent UI width. **Ctrl** plus **Up**, **down**, **Page Up/Down**, **Home**, and **End** can be used to scroll, as is common in the other text areas of the app. The **TAB** key will move to the next DAT file and then loop.

Alt-~ (Alt-Tilde)	Toggle persistent DAT peek. Regular game-list navigation changes the selection; **Ctrl** plus navigation keys scrolls the peek text.
Alt-1 through Alt-0	Go directly to art type. (left click on art area) Note: 'None' is always 0.
Alt-D	Cycle Games Only, Favorites & Games, and Favorites Only when favorites exist. With none, remain on Games Only.
Alt-Enter	Display properties for the game selected.
Alt-I	Cycle through the icon display sizing presets.
Alt-Insert / Delete	Cycles through Custom Game Lists (e.g. Full, Arcade, User Lists)
Alt-L	Save the current visible list as a new Custom Gamelist entry.
Alt-Left Arrow	Exits a DAT overlay, software list view, or clears an active filter.
Alt-O	Select a new random bkground.
Alt-P / TAB	Toggle through the different game list views. (right click on UI)
Alt-S	Cycles through the main 5 sort options while in flat view.
Arrow Left / Right	Navigate one letter's worth of games, from A to B to C, etc. through favorites and down through the main game list eventually looping to the top again.
Arrow Up / Down	Navigate one game at a time.
Ctrl-B	Benchmark the selected machine or software item with its actual host/media launch configuration; default is 90 emulated seconds.
Ctrl-D	Add or remove a favorite from favorites.ini.
Ctrl-Enter	Open the selected game's newest-first recording list, even when only one completed recording exists. Rows show date/time, filename or folder, and size in KB or MB; long filenames are shortened with an ellipsis. Works with Auto-Record Input off. In the list, **Enter** replays the highlighted session; **Backspace**, **Escape** or mouse Back returns to the game.
Ctrl-Delete (recording list)	Permanently delete the selected recording file and refresh the list automatically. Deleting the last session returns to the game; a failed deletion is reported.
Ctrl-F / Ctrl-E	Open text filter dialogue.
Ctrl-G	Edit Score, Date and Notes for the selected game. **Enter** saves; **Escape** or mouse Back cancels.
Ctrl-R	Select a random game.
Ctrl-Shift-0	Re-center and reset the dimensions to default without touching any other settings.
Ctrl-Shift-C	Copy the displayed results as tab-separated Name, Description, Year, Manufacturer and Software List columns. Machine rows leave Software List blank; software rows

	identify their canonical list. A short notice reports the copied count or a clipboard failure.
Ctrl-Shift-L	Cycle command line history in **F1**.
F1	Display the configuration dialogue.
F10	Internal tool for Build Verification Test.
F11	Toggles full screen mode, removes title bar and window frames.
F12	Show the grouped reference to the active shortcut bindings, including aliases.
F2	Launches the IV-Play.log overlay.
F3	Launches the IV-Play.cfg overlay.
F4	Link another MAME executable. An unsuccessful import or database load restores the previous link and setup and keeps the displayed list.
F5	Refresh the XML import from MAME.exe and rebuild icon / art caches.
F6	Run the art asset audit. If **F5** or another owned asset operation is in progress, **F6** waits and runs afterwards; pressing **F6** again during its own audit does not queue another.
F7	Toggle the onscreen diagnostic dashboard for FPS, memory usage for CPU and GPU, and garbage collection.
F8	After confirmation, Factory Reset removes IV/Play-owned generated files and resets cfg settings. It preserves unrelated files in a custom asset folder, along with Favorites.ini, IV-Play.Custom.ini, Hidden.ini, HiScore.ini and benchmark histories. If cleanup is incomplete, a warning names the problem rather than claiming success. Use **Shift-F12** if Available prompts for a new audit.
F9	Run the configured BigBench suite (default 36 games, 90 emulated seconds each), display its report and save a text report to the Desktop. Failed, invalid or too-short results are marked as errors and excluded from speed means and benchmark history. An unreadable or malformed prior history is reported as unavailable rather than as a first run. A new benchmark waits for a previous run that is still stopping.
Home / End	Move to the beginning / end of the game list, or as a stop at the beginning and end of the favorites (press **home/end** a second time to navigate out of favorites).
Ins / Del	Cycle through the dynamic themes while in the game list.
Page Up / Page Down	Navigate one window height worth of games.
Shift-1-9-Enter	Launches the selected game in N concurrent tiled windowed instances.
Shift-Alt-S	Cycle sort direction in Flat View.
Shift-Enter	Launch a softlist machine without entering its softlist. (effectively like powering on a console with no cartridge in it)

Shift-F8	Display a grid of all the IV/Play splash screens.
Shift-F12	Audit the linked emulator's rompath and select Available. The name-level scan considers ROM archives and CHD folders, including optional or undumped media rules; it does not inspect file contents. Wait for the scan to finish before relying on its results.

EXTERNAL FILES

IV/Play.exe can reside in its own folder or along with MAME.exe. It also now creates a sub-folder for its asset files for easy rebuilds. It's recommended to keep the *.inis at the same level as the .exe so they survive directory purges.

C:\games\IV-Play\IV-Play.exe (main application)
C:\games\IV-Play\IV-Play User Guide.pdf (this document)
C:\games\IV-Play\e_sqlite3.dll (required SQLite native library)
C:\games\IV-Play\Favorites.ini (user favorites list)
C:\games\IV-Play\IV-Play.Custom.ini (custom game list definitions)
C:\games\IV-Play\Hidden.ini (persistent machine/category/driver exclusions)
C:\games\IV-Play\IV-Play.cfg (application configuration)
C:\games\IV-Play\HiScore.ini (optional high score data for the art area overlay)
C:\games\IV-Play\IV-Play.Bench.History.txt (benchmark run history, one row per run)
C:\games\IV-Play\IV-Play.Bench.Machines.txt (benchmark history, one row per machine per run)

C:\games\IV-Play\IV-Play\IV-Play.available.bin (current results of **Shift-F12** available ROMs audit)
C:\games\IV-Play\IV-Play\IV-Play.background.state.json (background rotation tracking)
C:\games\IV-Play\IV-Play\IV-Play.db (SQLite game database)
C:\games\IV-Play\IV-Play\IV-Play.db.meta.json (database schema metadata)
C:\games\IV-Play\IV-Play\IV-Play.history.cache (parsed History.XML binary cache)
C:\games\IV-Play\IV-Play\IV-Play.icon.atlas.bin (raw binary icon texture atlas)
C:\games\IV-Play\IV-Play\IV-Play.icon.map.json (icon atlas coordinate index)
C:\games\IV-Play\IV-Play\IV-Play.icon.atlas.hash (atlas validity checksum)
C:\games\IV-Play\IV-Play\IV-Play.log.txt (diagnostic log)
C:\games\IV-Play\IV-Play\IV-Play.mameinfo.cache (parsed MAMEInfo.DAT binary cache)
C:\games\IV-Play\IV-Play\IV-Play.SoftListFavorites.cache (softlist favorites binary cache)
C:\games\IV-Play\IV-Play\IV-Play.SoftlistIndex.cache (140K-item softlist binary cache)
Benchmark cfg, snapshots and writable disk-difference copies now use owned temporary workspaces, removed after the run.

Once linked to a MAME.exe, IV/Play will use its structure for its art, for example:
C:\games\MAME\MAME.exe
C:\games\MAME\History.XML
C:\games\MAME\MAMEInfo.DAT
C:\games\MAME\INI\MAME.ini
C:\games\MAME\bkground*bkground*.png
C:\games\MAME\icons*gamename*.ico

C:\games\MAME\snap*gamename*.png
C:\games\MAME\flyers*gamename*.png
C:\games\MAME\etc\etc.png

IV-PLAY.CFG

This file is created on first launch of IV/Play if it is not present. It contains the configuration settings for the application. Delete this file (and the IV-Play sub-directory) to start fresh. Note that the *.cfg file can point to a custom asset directory (asset_directory), which is helpful if a RAM drive or similar technology is being employed for storage.

The following IV-Play.cfg options control native title-bar and window-border colours. Cfg edits made while IV/Play is closed take effect on the next launch. The enable toggle is also available in **F1**.

theme_titlebar=True (default) makes the title bars and outer window borders of IV/Play's native-framed windows follow the theme. Use Themed Window Borders turns this on or off in the Art and Filtering group, directly below Use Black Theme for Bright Backgrounds. Changes preview immediately; **Apply** or **OK** keeps them, and **Cancel** restores the opening or last applied choice. Set **theme_titlebar=False** to restore Windows' normal title-bar, title-text and border colours, respecting the System / Dark / Off mode. This includes the configuration window, Properties, Filter, Progress and the font/colour pickers. Their colours follow theme previews and return to the opening or last applied colours on **Cancel**. Windows file/folder pickers and message boxes keep their system appearance.

theme_titlebar_colour=3 (default) is cfg-only and selects the theme colour used for both the title-bar background and window border: **1** = artwork-border colour; **2** = main game-list text colour; **3** = scrollbar-thumb colour. The source follows the current static, dynamic or Custom theme as it changes. Other numbers use source 1.

IV-PLAY.DB

The SQLite database is created from the linked emulator's XML on first run, or rebuilt when the executable or required schema changes. IV-Play.db and its metadata file live in the configured asset directory (by default the IV-Play subdirectory beside the application). **F5** refreshes the linked executable's XML and caches; **F4** links a different executable. Both retain the Full or Available list and restore the selected game when it still exists. Required availability repair completes before the Available list is displayed. The matching e_sqlite3.dll must remain beside IV-Play.exe.

UPDATES

IV/PLAY 1.5 FEATURES

UNLIMITED ART TYPES

IV/Play supports multiple art types. In **F1**, add directories or archives containing images named gamename.png. On initial launch it probes for snap, flyers, cabinets, PCB, marquees, cpanel, titles, bosses, ends, gameover, logo, scores, select, versus, howto, warning, and artpreview, including their respective .zip or non-solid .7z packs. **F1**

allows the list to be reordered; startup and **F5** normalize it with Snap first. The selected art folder is retained through that normalization. Art shortcuts depend on the current order (**Alt-1** through **Alt-9**, with **Alt-0** for None).

HISTORY.XML / MAMEINFO.DAT SUPPORT

IV/Play supports History.XML and MAMEInfo.DAT. They are searched for in the MAME.exe directory on initial IV/Play launch and can be added at any time through the **F1** configuration menu. Activation of the view is via the **tilde ~** key which opens the DAT 'peek' view. The peek overlay is expanded left or right to multiple sizes and navigation is via the regular arrow keys, **page up** and **down**, and **Ctrl-End** and **Ctrl-Home**. A separate font with size and colour is selectable for the text area on the **F1** configuration menu.

FAVORITES TOGGLE

With favorites present, **Alt-D** cycles through Games Only, Favorites & Games, and Favorites Only, including software-list views and filter results. With zero total favorites, only Games Only is available; add a favorite with **Ctrl-D** to enable the other modes.

NON-WORKING GAME ART/HISTORY SUPPORT

IV/Play will display icons and any art type for non-working games if they are present. This also allows viewing of History.XML and MAMEInfo.DAT entries for non-working games.

COMMAND LINE OVERRIDE

The command line override in the **F1** configuration dialogue allows the use of various switches to be added to launched games, e.g. –window to play games windowed without having to drop to the mame.ini for editing. Note that command line arguments take precedence over MAME's *.ini processing hierarchy. The command line is now a drop down and has a well with the last 5 commands remembered. The command lines can also be cycled through with **Ctrl-Shift-L** and will complain if the command does not begin with MAME's expected dash (-) prepending the arguments.

IV/PLAY 1.6 FEATURES

Updated to work with new XML output of MAME .162.

IV/PLAY 1.7 FEATURES

CLONE AND NON-WORKING FILTERS

The **F1** clone and non-working display settings apply to the main game list and to software lists as soon as they open. Favorites stay pinned when these display settings hide an item from the main list.

IV/PLAY 1.8 FEATURES

- New background and modern Windows Segoe UI font as default on first run.

- **ESC** will exit IV/Play.
- It is now possible to un-indent the clones in **F1**.
- Relative paths are now used in the IV-Play.cfg file, this allows for easy portability/transport via thumb drive or removable storage.
- With 1.8.4.0 added support for the new format of the History file, now an *.XML at www.arcade-history.com.
- With 1.8.5.0 added the ability to change the colour of the non-working games/devices in **F1**.

IV/PLAY 2.0 FEATURES

After 9 years in maintenance mode, IV/Play sees a significant modernization effort, rebuild, major feature additions, and a full version bump in the summer/fall of 2025.

MOVE FROM .NET 4.5 TO .NET 8.0 & EXIT FROM WPFWRAPPER TO PURE WINFORM

A migration was undertaken successfully to get IV/Play from .NET 4.5 all the way to 8.0 and also to extricate it from the WPFWrapper which had ultimately caused bothersome flickering of the handoff on launch.

GPU RENDER PIPELINE

IV/Play has been converted from a GDI app to utilising a full GPU accelerated pipeline. This leads to considerably smoother scrolling and navigation in-app with art assets appearing instantly from its various caches. Art, icons and backgrounds scale via DirectX High Quality Bicubic instead of straight bilinear filtering, resulting in sharper images on higher resolution monitors.

This architectural change was a strategic decision to resolve a persistent and severe performance bug known as "scrolling judder". The previous GDI+ rendering pipeline was CPU-bound, meaning all drawing operations were handled by the main processor. During fast navigation, this would overwhelm the UI thread, causing the application to stutter or freeze.

The new pipeline offloads all rendering to the system's graphics card (GPU) using DirectX 11 / Direct2D. This frees the CPU from heavy drawing tasks, resulting in a perfectly smooth, zero-latency user experience, even when scrolling rapidly through thousands of games with high-resolution artwork. This one-time processing of assets on startup is an intentional trade-off that enables the application's highly responsive and fluid feel during use.

LITEDB TO SQLITE

IV/Play's data layer was migrated from LiteDB to SQLite during the modernization. The initial migration used Entity Framework Core; that layer has since been retired, and current database reads and writes use raw ADO.NET.

This was a critical architectural change driven by a series of persistent and difficult-to-diagnose crashes originating from the LiteDB query engine. The migration to a mature, industry-standard backend like SQLite completely resolved these stability issues.

SUPPORT FOR HIGH DPI MONITORS / LARGE ICON MODE / SCALING IMPROVEMENTS

- IV/Play is now high DPI aware; it won't be necessary to use Windows 11 compatibility hacks to enable a non-miniscule screen. The UI is now at 1523x648 which is proportionally what it was at 1015x432 on 1080p monitors. Running at 144 DPI and 4K now looks very nice; the fonts are crisp and each art asset looks the way it should instead of being simply stretched after a Windows snapshot. Everything scales properly based on the DPI setting including row height.
- Large icon view is now supported, it can be toggled in the **F1** dialogue and via **Alt-P** or **TAB** as one of the views in the cycle.
- Backgrounds will now scale themselves to the inner window size, a set size floor for that functionality can be altered in the *.cfg so anything smaller than the setting will be tiled. This works particularly well for aspect ratio friendly bkgrounds that may be slightly bigger or smaller in their dimensions than the current inner window.
- Long time bug fixed for maintaining a maximized state on re-launch.
- IV/Play no longer locks its assets with file handles, so the snaps viewed in that session can be replaced on disk underneath the app.

SUPPORT FOR SOFTWARE LIST DEVICES

IV/Play can now click/**Enter** on a console like Atari 2600 and it will read its softlist and display as the new game list. The navigation is the same as the full game list. **Backspace** or **Alt-Left-Arrow** leaves the software list and restores the preceding search results, with the system highlighted. In a filtered software list, press **ESC** once to clear the filter, then again to return to the previous results. The favorites work the same way in the softlist view, adding or removing them with **Ctrl-D**, or toggling their views with **Alt-D**. Use **Shift-Enter** to launch a softlist machine directly with no software/cart attached, effectively like powering on an Atari 2600 with no cartridge in the slot.

CUSTOM GAME LIST SUPPORT

Similar to the favorites.ini, IV/Play now allows the use of the IV-Play.Custom.ini file, selectable from the **F1** dialogue. If chosen, this file takes over the game list and displays entries contained within. This is a single file, but it can be nested based on [] or <> heading delimiters that will produce the drop-down choice. Heading names are case-sensitive, so [Shooters] and [shooters] are two separate lists. Full, Arcade, Available and Missing are the built-in lists; a heading with one of those names is skipped and noted in the log. If the file cannot be read (for example, an editor has it locked), IV/Play stays on the current list and shows a red "Can't read" notice.

#Custom GameLists
<Golden Age>
amidar
armora
astdelux
asteroid

<Fighters>
sf2
sfa3

GRID VIEW

Grid View displays games with descriptive labels under their icons. Navigation follows the game-list model. **Alt-P** or **TAB** cycles through Regular List, Large List, Flat View, Grid with art, and Grid without art; **Shift-TAB** reverses the cycle. If art is already None, the grid-with-art step is skipped.

BACKGROUND BRIGHTNESS DETECTION MODE

IV-Play can now automatically determine if the bkground image is of a very light/bright variety and adjust its fonts to black accordingly (which can still be overridden in the **F1** config).

IMAGE ATLAS FOR ICONS AND BINARY VERSIONS OF HISTORY.XML AND MAMEINFO.DAT

To maximize performance, IV/Play combines all individual game icons into a single large image file called an "icon atlas". This atlas is loaded into GPU memory on startup, allowing for extremely fast icon display and smooth scrolling by eliminating the need to load thousands of individual files from the disk.

To accelerate startup, IV/Play creates optimized cache files for History.XML and MAMEInfo.DAT the first time they are used. On all subsequent launches, the application reads from these pre-parsed files, dramatically reducing load times by avoiding the slow process of reading the large, original text files.

ADDITIONAL IMPROVEMENTS

- IV/Play.cfg is now alphabetized and sectionalized.
- Via the auto-record **F1** toggle, machines launched will have their session *.inp recorded. To play it back, launch the same game a second time with the **Ctrl** key held down.
- Nearest neighbor scaling is an option now in **F1** (unsmoothed snapshots); it allows a more pixelated scaling algorithm that lends a retro ‘air’ to the snaps.
- A simulated scanline effect can now be applied to snaps, both over the pixelated and the smooth versions.
- Clone icons and the focus/highlight are now transparent and controlled via settings in the *.cfg file.
- On a fresh 96-DPI desktop, IV/Play uses its historical 1015x432 dimensions and adjusts icon and row sizes. Current builds also rescale saved geometry when an installation moves between monitor DPIs; deleting the cfg is not required just to move from a 4K system to a 96-DPI display. **Ctrl-Shift-0** re-centers the window and restores default dimensions without changing other settings.

IV/PLAY 2.1 FEATURES

MOVE FROM .NET 8 TO .NET 9

Upgraded the project from .NET 8 to .NET 9 to benefit from the latest runtime performance enhancements, contributing to a more responsive UI and faster icon cache processing.

ADDITIONAL IMPROVEMENTS

- Tapping the **Tilde (~)** key will open the DAT 'peek' view where the DAT files are accessible in a temp window. **Ctrl** plus **Left** and **right** size the window to 40%, 75%, and full width of the UI. **TAB** cycles through the DATs.
- Copy filter results with **Ctrl-Shift-C** in tab-separated form for pasting into Excel.
- **F2** (log) and **F3** (config) now activate their 'peek' overlays with navigation the same as the DAT peek.
- **F8** performs a Factory Reset after confirmation: it resets IV-Play.cfg and removes named IV/Play-generated files. Favorites and other personal files are preserved; see the **F8** entry above for the current scope and incomplete-cleanup warning.
- Aspect ratio constrained sizing: If **Ctrl** is held down during a sizing event the UI will scale up or down in 2.35:1 TohoScope.
- **F7** toggles the performance overlay in the upper right corner, refreshed once per second while visible. It shows memory, handle, CPU and GPU readings, art-cache usage, garbage collection figures and draw timings. The one-core CPU figure can exceed 100% and does not identify UI-thread usage. Rendering is on demand, so idle FPS is not a performance target. Closing **F7** stops sampling; hiding IV/Play pauses it.
- Distributable is now packaged with AOT (Ahead-of-Time) compilation during publication (vs. JIT) which speeds the initial bring-up of the app.
- **F10** runs a short BVT (Build Verification Test) used in development to exercise basic functionality for breakage.
- Added a 'Theme' drop down to **F1** which allows a single switch for multiple colour combos on the various game list entries.
- Added ability to launch a softlist machine on its own, e.g. 'mame.exe a2600', instead of going into its softlist view. Hold the **Shift** key **down** while hitting **enter** or double clicking on the machine. This is akin to starting a console with no cart in it.
- Reintroduced an **F1** toggle to hide non-working mechanical machines, removing ~15K entries from the game list; this is now on by default.
- Added a drop down in **F1** for icon scale presets based on monitor resolutions and DPI combos. This controls the scaled size of the regular and large icon display on said monitors.
- **Alt-I** in the game list will cycle through the various icon scale presets.
- **F11** will now toggle into and out of full screen mode, hiding the window borders and title bar; similar to **F11** in browsers. **ESC** will also exit full screen mode. A default setting for this is user configurable in the *.cfg file in the window section.

IV/PLAY 2.2 FEATURES

VERTICAL SCROLLBAR FOR GAME LIST

A custom vertical scrollbar control has been added to the right of the art area.

FILTER IMPROVEMENTS AND ADDITIONS

The **Ctrl-F** filter has been overhauled with a clearer presentation and expanded functionality. Key features are now surfaced in the drop-down example list. Opening the filter on an empty field now shows a **Quick Start Filters** preview tier: 12 high-value curated filters shown immediately, with the full library accessible via **Alt-Down** or by clicking the hint line at the bottom of the list. Natural language terms such as *80s games*, *working only*, or *no clones* can be entered directly, while the full SQL-like advanced syntax remains available for precise control with

field-specific searches (e.g., year:1994), negation (manufacturer:!Capcom), and complex Boolean logic. New hybrid queries now allow natural language keywords to be combined with advanced operators in a single expression, enabling flexible searches such as Year >= "1995" noclones for parent games from 1995 or later, manufacturer:Capcom working for working Capcom titles, or description LIKE "%puzzle%" nonworking parents for non-working parent puzzle games. This hybrid approach bridges simplicity and precision, giving both casual and advanced users greater control over filtering results.

ADDITIONAL IMPROVEMENTS

- IV/Play's version information is now surfaced on the title bar after its name before the linked MAME .exe.
- The DAT Peek feature's key is now configurable; especially handy for non-US keyboard layouts. The default will try to find the key in the upper left above the **TAB** key and should handle most European keyboards. This key can be set in the IV-Play.cfg file. The various key codes may be found on Microsoft's site by searching for System.Windows.Forms.Keys enumeration in their .NET Framework docs.
- The **F1** config check boxes have been re-ordered and the verb has standardized on 'Show' vs. a mix of show and hide. Attendant logic has been updated for the new verbiage.
- The Grid View + No Art combo has been added to the **TAB** / **Alt-P** cycle as the fourth option.
- The diagnostic **F10** Build Verification Test (BVT) has been extended with a new overlay and progress screen with grid view interactions.

IV/PLAY 2.3 FEATURES

REINTRODUCED LEGACY ERROR DETECTION

Reintroduced the legacy IV/Play (1.8.5) feature that detects and reports MAME errors when attempting to run a machine without the required ROMs. This is displayed via one of the overlays and allows a copy and paste out of.

BIGBENCH BENCHMARK SCRIPT

Added John IV's *BigBench* benchmark script. Launchable via **F9** (and button on **F1** config), it runs 36 games in succession, measuring performance over 90 (emulated) second intervals. Monitoring overhead is negligible, and results are statistically equivalent to those obtained from a direct command-line run. Results are displayed in an overlay and also saved as a text file on the user's desktop. Both the game list and the -bench (time) value are configurable in the *.cfg file. See the benchmark page at the IV/Play site for historical data: https://john-iv.github.io/iv-play/bench.html. The geometric mean value is the current preferred measure of PC / MAME performance across disparate hardware.

DYNAMIC COLOUR THEMES

Added dynamic theme options in addition to the static presets accessible in the **F1** config. Dynamic themes calculate text and UI colours based on the dominant background image colour and its brightness. This allows the interface to adapt automatically to different backgrounds while maintaining legibility and acts on the fonts, scrollbar thumb, and art border. **Insert** and **Delete** cycle through the dynamic themes while in the game list.

- **Brightness Ladder** Adjusts text lightness relative to the background. On darker backgrounds, text colours are brightened in stepped increments to ensure readability. On lighter backgrounds, text colours are darkened to maintain contrast.

- **Saturation Gradient** Maintains a fixed lightness level but varies saturation. Favorites are rendered with higher saturation, clones with reduced saturation, and non-working entries in grayscale. This produces a consistent brightness level while distinguishing categories by colour intensity.

- **Analogous Palette** Selects colours close to the background's hue, shifted slightly on the colour wheel. A minimum lightness threshold is enforced to prevent text from becoming too dark. This produces a cohesive palette aligned with the background colour.

- **Dual Contrast** Applies both saturation and brightness adjustments to create stronger separation between categories. Favorites and parents are shifted lighter and more saturated, while clones are darker and less saturated. Non-working entries are rendered in grayscale with a minimum brightness floor to ensure visibility.

- **Miami Pastels** Selects a vibrant, high-saturation colour scheme using complementary pastel hues, typically teal and pink. Colours are assigned based on the background's general warmth (warm background triggers cool pastel text, and vice versa). A fixed lightness level is targeted to maintain the signature pastel aesthetic, with slight adjustments made for very bright backgrounds. This theme overrides the hue of the background completely to apply a distinct, stylized colour set.

EXPANDED KEY CUSTOMIZATION

Users can now modify several keyboard shortcuts to better suit their preferences. These customizations are made directly within the IV-Play.cfg configuration file. Look for the Key Bindings section near the top of the file; it includes comments explaining the syntax and providing examples of valid key combinations. Formalized the Euro keyboard version of the '**Tilde** Peek' key, which is **Oem8** and made it a standard alias.

FAVORITES.INI AUDIT

Toggled on by default in the *.cfg, this feature will audit the favorites.ini file on initial launch looking at each line. If it finds something it doesn't recognize it will prepend an asterisk to the entry; to surface it for manual editing and removal. It also displays an overlay with this information in-app. This feature helps keep track of occasional renames of machines that occur in baseline MAME .exe, and also clean up accidental misspelled manual entries to the favorites.ini.

ADDITIONAL IMPROVEMENTS

- The scrollbar thumb colour can now be customized in the **F1** configuration screen. The active scrollbar thumb colour may also be set manually in the *.cfg file. If left at the default, IV/Play will automatically adjust the thumb colour based on background brightness, selecting either a dark or light variant for optimal visibility. The scrollbar thumb also takes the dynamic themes colours if not user customized.
- The border opacity has been surfaced into a control on the **F1** config dialogue.

- Added a toggle for Random Dynamic Theme in **F1**. If checked, this will randomly cycle through the dynamic themes on launch, providing an attractive new look each session; especially when paired with randomizing the background.
- MAME and IV/Play versions now appear in the **F1** config form as well as a link to the home page.
- Benchmark button added to the **F1** config.
- The default short-cut keys will now display in the **F1** dialogue for easy reference.
- Added the ability to tweak which themes are available to the theme cycle and which are included in the randomization carousel in the *.cfg file.
- Added a button in **F1** to set the background images folder.
- **F6** launches an internal tool for icon and snap asset auditing.

IV/PLAY 2.4 FEATURES

SOFTWARE LIST FILTERING & FAVORITES

A large refactoring of the handling of software lists now allows for filtering/searching across the entire 100K plus collection of MAME software items. This means it's now possible to search for an item like name:Zaxxon and get the hits for Zaxxon across all of its ports. Software list items are also now eligible for inclusion to the favorites.ini file. Add or remove them with the same key **Ctrl-D**. They can be grouped at the end of the main game list favorites (sorted by machine) or integrated into the main list via descriptive name. The favorites audit will now extend to include these items, so a2600/combak vs. a2600/combat will be flagged.

F1 CUSTOMIZABLE KEYS SECTION

The main alterable key combos have been surfaced into the **F1** dialogue. All the remaining keys are editable in the IV-Play.cfg file.

THE ARCADE GAMELIST

IV/Play now has a curated, subjective, arcade-only game list; accessible via **Alt-INS** / **Alt-DEL** cycle keys or **F1** in the custom game list drop down. Excludes all the non-coin-op, mahjong, fruit, gambling, bar top, quiz, pinball, redemption, mechanical, and BIOS machines. With clones and non-working filtered out it's about 3400 games.

SUPPORT FOR ZIP AND .7Z ART PACKS

Added the ability to use \snap\Snap.zip|.7z and \icons\Icons.zip|.7z and the other art types, cabinets, flyers, marquees, etc. **Note**: 7z solid archives are not supported due to performance degradation, the app will warn if it finds one in its asset search. The Snap.7z and Icons.7z from the IV/Play Homepage can now simply be dropped in to the \icons and \snap directories for direct use.

MULTIPLE MAME LAUNCHES AND TILING

IV/Play now allows for using a single session of itself to launch multiple windowed instances of MAME; selecting either different games or the same game multiple times for multi-player gaming on a single screen. Use **Shift-1** through **Shift-9** while launching a game with **ENTER** or double-click to create that many windows. This allows for

playing networking supported games in MAME like linked Cruisn' USA cabinets or similar. **Note**: Only the first instance gets the sound, to prevent the cacophony of all of them outputting at once. The first windowed game also takes the mouse vs. the others for easier navigation and windowed placement, as desired.

MOVE FROM .NET 9 TO .NET 10

Upgraded the project from .NET 9 to .NET 10 to take advantage of long-term support and further runtime optimizations, delivering faster startup responsiveness, smoother UI interactions, and improved memory efficiency for a more stable user experience.

ADDITIONAL IMPROVEMENTS

- The art border size and the opacity settings now have a preview box next to them to visually balance out some of the other additions to **F1**.
- Verbose logging is now a toggleable option in **F1**.
- IV/Play's start-up time is now displayed on the **F1**, this includes time to an interactive UI and the final completion of background threads.
- Added a 5-layer history drop-down to the MAME command-line override combo box in **F1**, with the ability to assign a shortcut key to cycle, currently **Ctrl-Shift-L**, tweak in the *.cfg key mappings section.
- Added a 'User Guide' button to the **F1**, this will launch this document if it is in the same directory as the .exe.
- An individual machine can now be benchmarked; use **Ctrl-B** on a selected item to get the same result the full bench suite produces. The duration is the default 90 emulated seconds. This can be adjusted in the *.cfg file as desired.
- Allow a nonworking machine to display its own art asset if there is one.
- Prevent snap scanlines on vector games and non-MAME generated art (flyers, marquees, PCBs, etc.)
- Allow saving off the random bkground tracking json file before a full factory reset.
- Allow DAT peek with art type 'None'.
- Auto-populate paths for the current common art types if present in the MAME.exe directory structure.
- Added Record INPs and Launch in Fullscreen as toggles in **F1**.
- The app now hides itself when launching MAME.
- Reworked the status line on the game Properties view (**Alt-Enter**) to provide more detailed and clearer information presentation.
- The **F4** executable picker defaults to all files, making differently named MAME/HBMAME builds easier to link. IV/Play's supported platform remains native Windows.
- Added a toggle in **F1** to show or hide Softlist machines themselves.
- Added ability to search by softlist media types like cartridge, cassette, CD-ROM, etc.
- Added autocomplete with ghost-text look-ahead to the **Ctrl-F** filter dialogue. See the 2.16 section for the current history, completion and keyboard improvements.

IV/PLAY 2.6 FEATURES

FLAT VIEW & GLOBAL SORT

Added a new flat view which displays all machines alphabetically and non-indented; clones decoupled from their parents, favorites not pinned, parents at full opacity. Accessible via the **TAB** cycle or the **F1** View dropdown. Flat View is also the only mode that supports global sort, which orders the list by default, description, machine name, manufacturer, or year. The selected sort field and direction persist through restarts. **Left/Right** follow the chosen sort: title or short-name initials, years, or manufacturers. Sort cycles via **Alt-S** or the **F1** Sort dropdown; especially useful when a filter result contains thousands of entries.

DEEPER DAT INTEGRATION FOR FILTER SEARCHES

History.xml, MAMEInfo.dat and CatList.ini participate in **Ctrl-F** searches. history: / hist: searches History.xml text for machines and software; mameinfo: / mi: searches the full MAMEInfo.dat text for machines, while mameversion: / mv: selects the first-appearance version. genre: / gg: searches CatList.ini categories and subgenres. These sources also support natural-language searches such as imperfect shooters from 1995. workingstatus: and supportstatus: distinguish machine driver status and software support level. See Appendix 1 for syntax and examples.

AVAILABLE ROMS GAMELIST

The Available gamelist, selected in **F1** or via **Alt-Insert/Delete**, audits .zip/.7z names on the linked emulator's rompath and shows only the machines it finds. Detection is name-only. The list intentionally does not show ROM-less games like Pong; switch back to the full gamelist to see those entries. **Shift-F12** builds the audit. See Game List above for how games are matched.

PLATFORM-BASED SOFTLIST FILTER GROUPING

When a filter (**Ctrl-F**) returns software list results, IV/Play now collapses them into platform-level family nodes rather than producing one row per software item per hardware variant. A search for 'zaxxon' produces a single 'Zaxxon (Atari 2600)' entry, a single 'Zaxxon (ColecoVision)' entry, a single 'Zaxxon (MSX1)' entry, and so on; one row per platform, regardless of how many hardware machine variants or regional releases exist for that platform in MAME. Pressing **Enter** on a family node drills down into it, showing the parent release and any clones (regional variants, alternate revisions) for that platform. The parent entry shows its clone count, and each clone shows its parent name in brackets. Press **Alt-Left Arrow**, **Backspace**, or **ESC** to return to the platform-level filter results. Adding 'noclones' to the filter suppresses clone entries before grouping runs, so only parent releases appear and drill-down shows a single launchable entry.

NATIVE AOT TRANSITION / PERFORMANCE

IV/Play ships as a fully self-contained Native AOT executable and does not require a separate .NET 10 runtime installation. Native code generation avoids runtime JIT compilation during ordinary use; the distributable includes the required runtime support. The SQLite native library remains a separate e_sqlite3.dll beside the executable.

MACHINE STATUS AWARE ICON BORDERS

This configuration feature surfaces the status of MAME machines that appear in **Alt-Enter** properties as not working (red), imperfect (orange), and working (green) by using the black border of the existing icons and changing

them to that colour (or any user chosen in the *.cfg). Further, since the imperfect games are the most in need of visual differentiation, they are now defaulted to a bkground derived themed colour to complement the auto-theme feature. This auto colour for the imperfect status border can be turned off in **F1** to return to user chosen RGB settings. It also extends to softlist machine media items.

HIDDEN.INI

IV/Play now supports a hidden.ini file at the executable level for permanently suppressing entries from the game list. Each line takes one of four prefix forms: type: targets a machine category, machine: targets a specific short name, driver: targets all machines sharing a source driver, and manufacturer: targets every entry whose manufacturer contains the given text. Excluded entries do not appear in filter results, favorites, or any custom gamelist. The file survives factory resets. It is generated automatically if not present and contains the syntax/usage information; its creation can be suppressed by a toggle in the *.cfg.

CUSTOM GAME LIST GENERATION FROM FILTER RESULTS

Using **Alt-L** from a filter result set will now automatically create an entry in the custom-list.ini file.

NUMBER OF CLONES / PARENT

Added the ability to show the number of clones a machine has on the gamelist, as well as the counterpart `[parent: x]` on clone entries. Clone counts use singular/plural forms (“1 clone” / “5 clones”) and appear left-anchored immediately after the machine description. This can reveal interesting historical patterns where widely bootlegged games may have dozens of clones. Accessible in **F1** as two independent toggles.

COMMAND LINE OVERRIDE AUTOCOMPLETE

Returning from IV/Play 1.8, MAME’s `-showusage` output is captured and used for the command line override text input.

ADDITIONAL IMPROVEMENTS

- DPI detection and display has been hardened; IV/Play now flows better when dragged across multiple monitors with different DPIs and scaling settings.
- Added toggles for ‘Show Parents’ and ‘Show Machines’ to create even more display permutations. These can also be accessed via natural language queries in **Ctrl-F** such as “Parents” or “Machines Only.”
- Added new randomized splash screens which appear on full MAME XML export and .db rebuilds at launch.
- Added **Shift-F8** to show all the new internal splash screens in an overlay panel.
- Added two new 3-way filter operators to **Ctrl-F**: workingstatus:working / :imperfect / :notworking (alias ws:) distinguishes fully-working from imperfect-emulation machines, going beyond the boolean working only phrase. supportstatus:yes / :partial / :no (alias ss:) filters softlist items by their MAME support level. Combine with explicit system scope for best results: system:snes supportstatus:yes. Natural language aliases such as imperfect machines, broken games, and partially supported are also supported.

- Added a ‘snapshot’ of the entire user configuration taken on launch and available in the **F2** log. If changes are made through the session, they are flagged by a colour change and delta icon (Δ) on the next **F2** to make for easier debugging/diagnostic work.
- Added Genre, MAME version, and machine type to **Alt-Enter** properties since we consume all of that and hydrate it now. Requires History.xml, MAMEINFO.dat, and CatList.ini.
- Added a subtle configuration/**F1** button at the bottom of the vertical scrollbar. This is the first time in the app’s history that the configuration dialogue is accessible without a keypress.
- Added current game list in use to the titlebar.
- Add **Alt-O** to select a new random bkground. **Note** shortcut keys are all assignable in the *.cfg and some major ones in the **F1** dialogue.
- Rename favorites and custom gamelist in the distribution packages so they don’t overwrite users’ own files.
- Added the ability to return the app to center of screen and its default dimensions. Double click the titlebar or use **Ctrl-Shift-0**.
- Added an OS aware System / Dark / Off mode drop-down in **F1**.
- Added a browser-like **Ctrl-F** (Find) inside the overlays, including **F2** & **F3**, and the DAT peeks.
- Modernized the font and colour pickers in **F1**.
- Removed hardcoded Segoe UI usage throughout the rest of the app, main UI font now set in the **F1** config and will propagate. New default per Windows 11 is Segoe UI Variable Text.
- Created a TTF font based on MAME’s very old uismall.bdf for use in MAME’s internal UI. Double-click it, choose ‘Install’ to bring it into Windows 11, then use ‘-uifont MAMEUISmallMod’ in the MAME.ini to utilise it, in-game. It also looks nice as the UI font in IV/Play itself, set in **F1**. Added to the distribution package.
- Added a third font picker field in **F1**, this for the rest of the UI, including dialogues. This allows a separate gamelist font setting to be decoupled and individualized.
- No longer including favorites.ini and the custom gamelist.ini in the distribution package so there is no accidental overwriting on decompressing the archive.

IV/PLAY 2.8 FEATURES

DPI & LAYOUT MODERNIZATION

IV/Play’s DPI system has been rebuilt to eliminate the mix of pixel-based and DIP-based sizing that earlier versions relied on. All layout calculations now flow through a unified DPI-aware geometry model, so the interface scales cleanly on high-resolution displays and behaves correctly when moved between monitors with different scale factors. Window metrics, icon sizes, and hit-testing now update deterministically when DPI changes, preventing the fractional drift and inconsistent sizing that could occur before. The result is a sharper, more stable presentation on modern 4K and mixed-DPI setups without requiring any configuration changes from the user.

INTERNAL HISCORE.INI

Hiscore.ini which sits beside IV-Play.exe stores personal score text and optional dates for the artwork annotation. Machine keys are plain shortnames (pacman); software uses its software-list name and item (a2600/pacman or adam_cass/buckrog), matching Favorites.ini even when the launch host differs. A software item never borrows a same-named arcade score. Position and opacity are configurable in the cfg, and **F8** preserves the file. The current

five-column format, **Ctrl-G** editor and catalog review are described under IV/Play 2.16 Features below. External edits still require a restart for display unless a successful editor save republishes the file.

The Hiscore opacity setting now defaults to 100%. On upgrade, a saved 50%, the previous default, is changed to 100% once; other configured values are retained. A later deliberate 50% setting remains available through hiscore_opacity in IV-Play.cfg.

MAME SOURCE .CPP ADDED TO DAT PEEK

Added a link to the MAME source to the DAT Peek feature. The DAT peek cycle will include the machine in question's XX.cpp file so it is easily accessible for informational purposes. MAME source path is set in the *.cfg source_root line.

UNATTENDED BENCHMARKING (-BENCH)

Launching IV-Play.exe with -bench runs the configured suite without opening the UI, writes a report and exits. Games and duration come from IV-Play.cfg, as for **F9**. Eligible runs append IV-Play.Bench.History.txt and IV-Play.Bench.Machines.txt beside the executable; these survive **F8** and open in Excel. See Reading Benchmark Reports and Histories in the Technical Companion for interpretation and exit codes.

ADDITIONAL IMPROVEMENTS

- Added TohoScope Internal / External to the *.cfg (and **F1**). This is an amusing little feature that will alter the dimensions of the main UI to conform with TohoScope's 2.35:1 aspect ratio in two different ways. The current default method and the one that has been used since IV/Play's inception is 2.35:1 for the *entire* app including the Windows chrome. That always meant the internal window area was not TohoScope as a result. The new setting 'Internal' will now make the inner window area 2.35:1, increasing the total height of the app by the titlebar size.
- Live ROM availability monitoring via FileWatcherService. When a ROM is added or removed from a configured ROM directory, the Available gamelist updates automatically approximately one second after the file settles, without requiring an **Shift-F12** manual scan.
- Auto-refresh of favorites sync on external favorites.ini changes, also via the FileWatcher pipeline.
- Added a check for update button on **F1**, this will ping the website and compare the local .exe version's GUID against the release on the web site and provide a link to the releases page to download the newer one.
- Introduced a 'persistent' DAT view. Toggle with the **Alt-Tilde** key and control the interior navigation of the DAT overlay with **Ctrl** key combinations, e.g. **Ctrl-Down**, **Ctrl-Page Up**, **Ctrl-Home**, etc. Regular navigation keys will move through the gamelist, selecting a new machine. Note: Regular **tilde** key launches the regular DAT peek which allows full control
- Fixed window-restore under multiple simultaneous MAME launches. IV/Play now stays hidden until the last running instance exits, rather than reappearing as soon as the first one closes.
- Refined CHD filters. chd games now matches any arcade machine that requires a CHD image, and a new chd software filter isolates CD-ROM and hard-disk software list titles (see Appendix 1).
- Screen-count filters are now exact. 2 screens, 3 screens, and 4 screens match machines with precisely that many displays so results line up with MAME's own totals; multi-screen still means two or more.

- History: / hist: searches now include software list items, not just arcade machines - the DAT text for software was always present and is now reachable from the filter bar.

IV/PLAY 2.12 FEATURES

- HBMAME support improvements. IV/Play reads hbmame.ini for HBMAME, identifies it in the title bar and **F1** About, and lets **F4** link differently named emulator executables.
- **F1** now fits smaller work areas with scrollable settings and a fixed footer, applies newly selected icon sources on OK, and keeps the highlighted game when it remains available. Art selection follows its folder through sorting and removal instead of drifting with a list position.
- Steadier DAT peek rendering. Navigating to a new game while the peek is open no longer briefly blanks the text, and the media-type column now reads consistently. Removed filter-term highlighting inside the DAT peek. It added little once the peek gained its own in-text find (**Ctrl-F**), and the two were unrelated.
- The mouse Back button closes overlays, such as the DAT peek, the **F2** log, the **F3** cfg and the benchmark window, as **ESC** does.
- SQLite updated to 3.53.4. When upgrading, replace e_sqlite3.dll together with IV-Play.exe; the **F2** log names the SQLite version actually loaded.

IV/PLAY 2.14 FEATURES

- Refined Arcade Only list. Restored video games that were unintentionally excluded and improved filtering of gambling, medal, quiz and photo-booth machines. Inclusion and exclusion exceptions now handle related clones more consistently.
- More consistent Arcade results at startup. Category information from catlist.ini is loaded before the Arcade list is filtered, and loading is retried if the file was not yet available.
- More accurate Available list. Games that need a CHD are listed only when their CHD folder is present, and in a merged collection a game is listed only when its BIOS archive is present too. Boards whose ROMs all come from their BIOS, such as the Neo Geo MV-1, are listed from the BIOS archive, and in a split collection a clone needs its own archive. A ROM or disc that MAME marks as never dumped is not required. Detection still reads names only, never file contents. When IV/Play cannot tell how a collection is organized, it lists the game rather than demanding a separate BIOS archive.

IV/PLAY 2.16 FEATURES

SEARCH AND RECENT SEARCHES

Ctrl-F now remembers 20 distinct searches, including ones that found nothing; older ones are under More recent searches… in the drop-down.

TAB accepts a suggestion or moves to the buttons, **Shift-TAB** moves backward, and **Home/End** now edit the text instead of jumping through the popup. An invalid query shows one error and keeps the current results; mistyped fields are named.

After a main-list search finds no matches, reopening **Ctrl-F** can offer a grey system-scope suggestion, such as "→ Try: system:a2600 dkong [**TAB**]" for dkong a2600. **TAB** adopts it, **Enter** runs it, **Ctrl-Z** restores your original text and **Escape** dismisses it.

Searches also follow the displayed list more consistently: noclones narrows that list, while title text or an explicit software request brings software-list results into Full. **Ctrl-Shift-C** now adds a Software List column and reports the copied count or a clipboard failure.

SOFTWARE LISTS AND NAVIGATION

Games Only, Favorites & Games and Favorites Only now behave the same inside software lists. Leaving a software list returns to the search and system that led into it; the first **Escape** clears the list's own filter and the next one leaves. Software items keep their own identity for favorites, artwork, recordings, benchmarks and copied results.

Flat View keeps its sort field and direction after restart, and **Left/Right** follow the chosen order. In grouped views, letter jumps now include clones whose parents are absent.

List and Flat title ordering now place an original before its numbered sequels, such as Marvel Vs. Capcom before Marvel Vs. Capcom 2.

Software families now select compatible launch hosts for each release and keep different media families separate. Explicit system scopes match the requested system rather than an accidental name prefix.

SETTINGS AND DISPLAY

F1 Apply now adopts changed views, sorting and art, icon or DAT sources immediately, and **Cancel** restores the opening choices or the last **Apply**. Colour swatches work from the keyboard, leaving fullscreen keeps the current monitor, and changing TohoScope updates the window shape immediately.

Dynamic themes now check their text colours against the background for readability, and background sampling follows the visible crop or tiling as the window resizes. Miami's bright-background adjustment is restored.

MAME LINKS AND AVAILABLE GAMES

An unsuccessful **F4** import now restores the previous MAME link and setup. A copied folder containing both IV/Play and MAME uses the copy's own MAME. Refresh can retry an unavailable software index without restarting IV/Play.

Available now follows the emulator's configured paths more accurately, including quoted paths, repeated INI settings and environment variables. Optional and undumped media no longer hide an otherwise eligible game, and live updates now include software-list archives and CHD folders.

HISCORE.INI EDITING AND CATALOG AUDIT

Press **Ctrl-G** on a selected machine or software item to edit Score, Date and Notes in a compact overlay. All three are free text. **Enter** saves and closes; **Escape**, mouse Back or moving focus away cancels.

Hiscore.ini now has a header and five tab-separated columns for spreadsheet use: Name, Score, Date, Full Name and Notes. Older files are converted when loaded, saving won't overwrite a record changed since the editor opened, and **F8** preserves the file.

After launch and after a catalog refresh, an audit updates known descriptions. Missing keys open a review: choose a suggestion or type a valid key and press **Enter**, or leave the entry unchanged. Nothing is renamed automatically.

RECORDED INP SESSIONS

Auto-Record now keeps every session under a unique dated AM/PM filename. **Ctrl-Enter** opens the game's recording history, newest first, even when Auto-Record is off. **Enter** replays a session and **Ctrl-Delete** permanently removes it. Playback depends on a compatible MAME build and game state; IV/Play does not manage those prerequisites.

BENCHMARK RESULTS AND UNATTENDED RUNS

Only usable, sufficiently complete results now count toward speed means and histories; failed or short runs keep their diagnostics. Reports now tell a damaged history from a first run.

The full report's AVERAGE row now shows total measured game wall time, in seconds and in m:ss.

ARTWORK REFRESH AND ICON SCALING

Artwork now stays tied to the selected folder or archive, even when separate sources share a name. Missing art falls back to another configured Snap source. Software-list icons and multi-size icon files display more consistently.

Artwork and icon archives can now be replaced while IV/Play stays open; press **F5** afterwards to pick up the changes.

During rapid navigation the previous image now stays up until the next one is ready. Snapshot scanlines scale with the displayed artwork. Icon sizes now scale a fixed master on the GPU, so size or DPI changes no longer rebuild the atlas. The old Force 64 option is removed.

KEYBOARD REFERENCE AND PERFORMANCE DASHBOARD

A grouped, scrollable shortcut reference shows the current assignments, including aliases. **F7** now keeps a stable panel width as values change. **Escape**, mouse Back or the overlay's own shortcut closes either one.

ADDITIONAL IMPROVEMENTS

- Long notices wrap inside the window.
- Factory Reset now preserves unrelated files in custom asset folders and reports incomplete cleanup. Unicode paths and saved fonts work more consistently across Windows regional settings.
- DAT text and in-text Find follow the selected game and source, and path changes in **F1** take effect without a restart.

IV/PLAY 2.17 FEATURES

MOUSE BACK AND FORWARD

Mouse Back now leaves recording history for the same game, dismisses overlays and retreats through family, software-list and search layers, one step at a time.

Mouse Forward opens the selected family or software list. After leaving recording history with Back, Forward reopens that game's recordings with a fresh file scan. Side-button navigation does not launch a game or replay a recording.

Back over the visible game list leaves a running benchmark alone; Back over the benchmark window stops it. A side-button release is ignored if the list rebuilt while the button was held.

RECORDING HISTORY

Ctrl-Enter now opens recording history even for a single completed INP, so you can inspect it before playback. A new right-aligned size column shows KB or MB; long filenames shorten with an ellipsis.

SETTINGS AND DISPLAY

F12 now opens shortcut help; **Shift-F12** rebuilds Available. Previous default assignments migrate automatically, while other custom bindings are kept.

Windows 11 title bars and outer borders now follow the current theme, including **F1** and sibling dialogs. **F1**'s Use Themed Window Borders toggle previews the change; Cancel restores the opening or last applied choice.

Hover over the selected list row's clipped description to read its complete text.

Hiscore artwork text now defaults to 100% opacity. A saved 50% from the previous default moves to 100% once; other saved values are kept.

Theme-coloured notices choose contrasting light or dark text against the displayed artwork border or wallpaper. Game icons keep their correct images when moving between monitors with different DPI settings.

AVAILABLE GAMES AND BENCHMARKS

Available now catches changes made while monitoring was stopped for a refresh or rebuild. Completion counts match the displayed results, automatic rescans notify only when availability changes, and the latest notice waits until you return from a game.

Default -bench runs now skip a MAME build already recorded for the same host, CPU and launch path; use -bench -bench-repeat for an intentional repeat. **F9** and **Ctrl-B** remain available for manual comparisons. A duplicate skip exits with code 8; unavailable admission evidence or an invalid machine roster exits with code 9.

Benchmarks use temporary cfg and snapshot folders and copies of existing disk-difference state, preserving the originals and configured hash/sample paths. **Ctrl-B** now includes the MAME build timestamp and host/hardware summary, with a larger, wrapping results window.

ADDITIONAL IMPROVEMENTS

- Hiscore.ini now preserves ANSI characters from spreadsheet or hand edits and writes UTF-8 on the next save. Undecodable files are retained.
- From a focused permanent History or MAMEInfo text view, **Escape**, **Backspace**, **TAB** or **Shift-TAB** returns to the game list. A **Shift**-number multi-launch choice clears when **Shift** is released or focus leaves the list.
- Artwork refresh and rapid navigation no longer reuse stale pictures or apply another selection's image dimensions. If a new MAME database fails to load during **F4**, IV/Play restores the previous link and database together.
- The executable's Windows File description now exposes its build GUID for comparison with **F1** About.

APPENDIX 1 - FILTER & SEARCH GUIDE

The filter bar (**Ctrl-F** or **Ctrl-E**) accepts plain English phrases, keyword modifiers, and a full SQL-like advanced syntax. All three modes can be freely mixed in a single query.

SIMPLE TEXT SEARCH

Type any word or phrase. Multiple terms side by side are combined with AND. Quoted phrases match anywhere in description, name or manufacturer, ignoring case: "king of fighters" includes The King of Fighters 2002, and "donkey kong" includes its sequels. Words inside quotes remain literal text; explicit description = "Donkey Kong (US, Set 1)" instead requires the whole stored field. Words in a title stay title words, even beside a field expression: lost in space, some like it hot, dead or alive and crime does not pay search their title words. All word operators require uppercase AND, OR, NOT, IN, LIKE and BETWEEN; field names and search values still ignore case. Natural-language phrases such as not working, between 1991 and 1995 and game and watch retain their meaning. A bare four-digit number matches the year or the title, so 1943 also finds 1943: The Battle of Midway. A one-word genre, machine-type or input keyword (golf, pinball, mouse, lightgun, cocktail) also matches titles containing the word, so mouse trap finds Mouse Trap; the phrase forms (golf games, pinball machines) are the keyword alone.

pacman midway	Matches 'pacman' and 'midway' in description, name or manufacturer; an unquoted word can also match an exact machine driver filename stem
"metal slug"	Phrase-anywhere match in description, name or manufacturer; quotes keep the words literal and omit driver matching
capcom fighting	All Capcom fighting games (both words must appear)
system:snes Zelda	Explicit SNES scope + keyword - finds Zelda titles for SNES

SYSTEM NAME ROUTING

Ordinary unquoted words search description, name and manufacturer, and also match the exact driver filename before .cpp or the ROM Set Of name, ignoring case. model2 finds machines in model2.cpp; galaxian includes other games in galaxian.cpp, and neogeo includes machines in neogeo.cpp. Folder names and partial filenames do not match: nes does not match snes.cpp. For example, naomi2 also finds games whose ROM Set Of is naomi2. This matches the direct ROM parent, without following clone ancestry. Quotes omit driver and ROM-parent matching: "galaxian" searches text only. Software never inherits its host driver or ROM parent. Use romof:naomi2 to search the ROM Set Of field directly. bios alone selects BIOS entries; bios: is unsupported and reports a Filter Error with a romof: hint. Existing year/date handling and visibility filters still apply. Bare system shortnames never activate platform scope; use system:<shortname> to select a machine together with its software, or add a routing qualifier to select a specific tier.

system:nes	Both the NES hardware driver AND all NES softlist media (explicit system scope)
nes machine / nes hardware / nes system	Only the NES hardware driver entry (Tier 1). Qualifiers: machine, hardware, system, platform, driver, host
nes games / nes titles / nes software	Only NES softlist media - no hardware driver (Tier 3). Qualifiers: games, titles, media, software, items, library, catalog, collection
nes cartridges / nes floppies / nes cdroms	NES softlist media filtered by media type. Any media-type word (cartridges, floppies, cdroms, cassettes, hdds, memcards) triggers this rule
softlist systems / softlist devices	All hardware drivers that expose a software list, across all platforms (Tier 2)
software only / media only / softlist	All softlist media across every platform, no hardware drivers (Tier 3)
machines only / no softlist	All hardware drivers only, no softlist media (Tier 1)
arcade media	ROM-based arcade machines (MediaType = rom). Distinct from arcade only, which uses the IsArcade flag

Note: qualified forms remain concise: nes machine selects hardware only, nes games selects softlist media only, and nes cartridges adds a media-type restriction. For arbitrary combinations, use explicit scope - for example, system:snes zelda or system:snes supportstatus:yes.

A failed plain main-list search can offer an explicit system-scope alternative when **Ctrl-F** is reopened. For dkong a2600, the grey hint offers system:a2600 dkong. **TAB** adopts it for review, **Enter** submits it and **Ctrl-Z** restores the original. This optional suggestion does not change the meaning or results of the original search.

KEYWORD MODIFIERS

Add these keywords to any search - or use them alone. Keywords can be combined freely.

WORKING / CLONE / PARENT STATE

working only	Show entries not marked non-working / unsupported (broad IsWorking filter)
nonworking / nonworking only	Show only non-working games
imperfect	Show games with 'imperfect' driver status
parents only	Hide all clone games
no clones / noclones	Hide all clone games (aliases)
clones only	Show only clone games (overrides global hide setting)
not parents / no parents / nonparents	Show only clone entries (inverse of parents only)
working parents	Working non-clone games only
nonworking parents	Non-working non-clone games only
working clones	Working clone games only
nonworking clones	Non-working clone games only
working clones of nonworking parents	Playable clones whose parent set is broken

nonworking parents with working clones	Broken parents that have at least one working clone

MACHINE WORKING STATUS (3-WAY)

Use workingstatus: (alias: ws:) to distinguish the three MAME machine driver statuses precisely. working only and nonworking use the broader IsWorking boolean: good and imperfect machines are both working; for softlist items, support yes and partial are both working while no is not. Use workingstatus: to isolate imperfect from fully good machines.

workingstatus:working / ws:working	Fully working only - DriverStatus = “good”. Also: fully working games, fully working machines
workingstatus:imperfect / ws:imperfect	Imperfect emulation only - DriverStatus = “imperfect”. Also: imperfect machines, imperfect games
workingstatus:notworking / ws:notworking	Not working / preliminary - DriverStatus = “preliminary”. Also: broken machines, broken games, not working machines, preliminary

CONTENT TYPE

softlist / software / software only	Show only software list items (console/computer games)
machines only / no softlist	Show only arcade/computer machines, no software items
softlist devices / softlist systems	Machines that have a software list attached
arcade only / coin-op / coinop / coin op / arcade games	Coin-op, non-mechanical, non-redemption video games only
non-arcade	Everything except coin-op arcade machines
bios only	Show only BIOS entries
no bios	Hide all BIOS entries

mechanical	Show mechanical games (pachinko, pinball, etc.)
working mechanical	Working mechanical games only
not mechanical	Exclude mechanical games
devices / device only	Show only MAME device entries (sub-boards, BIOS helpers - not launchable)
no devices / exclude devices	Hide all MAME device entries from results
console / home consoles	Home console hardware (machinetype:console)
computer / home computers	Home computer hardware (machinetype:computer)

FAVORITES

favorites only	Show only games in the favorites list
exclude favorites	Hide all favorites from results
not favorites	Alias for exclude favorites
nofavoriteslist / no favoriteslist / noflist	Suppress the pinned Favorites group at the top of the list without removing favorites from results - creates a single flat list. Useful with broad filters: softlist noflist, capcom noflist

HARDWARE & DISPLAY

vector	Vector graphics games (Asteroids, Tempest, etc.)
raster	Raster (pixel-based) graphics games
vertical / portrait	Vertically oriented monitor games

horizontal / landscape	Horizontally oriented monitor games
chd games	Arcade/PCB machines that require a CHD image - any machine with a disk in its manifest (CD-ROM, laserdisc, or hard disk)
chd software	Software list titles on CD-ROM or hard-disk media - the software-side counterpart to chd games
laserdisc games	Machines that use laserdisc media
cd-based	Arcade/PCB machines with CD-ROM hardware ⚠ Hardware filter - NOT the same as 'cdroms' which filters softlist CD media
samples / no samples	Show or hide games requiring external sound sample files
screenless	Machines with no display output
1 screen / one screen	Machines with exactly one display
2 screens / 3 screens / 4 screens	Machines with exactly that many displays. Word forms such as “two screens” and “N monitors” also work
multi screen / multiscreen	Machines with two or more displays
no protection	Hide games with emulated protection schemes
not bootleg / no bootlegs	Hide bootleg versions
cocktail / cocktail games	Machines supporting cocktail / flip-screen mode (HasCocktailDip = true)
no cocktail / upright only	Machines that do NOT support cocktail mode (HasCocktailDip = false)

cpu:z80	Machines using a specific CPU chip. Substring match on the chip list - cpu:z80, cpu:68000, cpu:arm. Multi-word names work: cpu:zilog z80. Machines only - not available for softlist items.
sound:ym2151	Machines using a specific sound chip. Substring match on the audio chip list - sound:ym2151, sound:oki, sound:pokey. Machines only - not available for softlist items.

CONTROLS

lightgun / lightgun games / light gun	Show lightgun games
no lightgun / no light gun	Hide lightgun games
trackball / dial / paddle	Games using these specific analog controls
dual stick / dual joystick	Games using a dual joystick setup
joystick	Games using any joystick input
2-way / 4-way / 8-way	Games with specific joystick directions
spinner / spinner games	Games using a spinner (rotary) control
mouse / mouse games	Games using a mouse or trackpad control
mahjong / mahjong games	Mahjong games (mahjong-keyboard input)
keyboard games	Games using a keyboard as primary input
keypad	Games using a numeric keypad
gambling games	Games using gambling-style input hardware

no buttons / 0 buttons	Games with no buttons
1 button / one button	Games with exactly 1 button
2 buttons ... 6 buttons	Games with 2 to 6 buttons (word forms also work: 'six buttons')
1 player ... 4 player	Filter by player count
multiplayer / co-op / 2p	Games supporting 2 or more players (also: 3p, 4p)
1p / singleplayer	Single-player games only

MEDIA TYPE (SOFTWARE LIST ITEMS)

These filter software list items by physical media type. For arcade CD hardware, use cd-based or chd games instead.

cartridges / cart games / modules	Cartridge-based software (media:cartridge)
cassettes / tape games	Cassette tape software (media:cassette)
floppies / floppy disks / diskettes	Floppy disk software (media:floppy)
cdroms / discs / disc games / cd-rom games / optical discs	CD-ROM software (media:cdrom)
hard disks / hard disk / hdds / hdd / hdd games	Hard disk software (media:harddisk)
memory cards / memory card / memcards / memcard	Memory card software (media:memcard)

MACHINE TYPE

Natural language phrases for machine type categories (requires history.xml in Art Paths). The machinetype: operator (aliases: mtype:, mt:) performs a substring match against the machine-type descriptor extracted from

history.xml. Some NLP keywords route via genre: (catlist.ini) instead where no history.xml descriptor exists - noted below.

fruit machines	Fruit machines / AWPs (machinetype:fruit machine)
slot machines / slots	Video and mechanical slot machines (machinetype:slot machine)
pinball	Pinball machines (machinetype:pinball), plus titles containing pinball; pinball machines is the type alone
electro-mechanical / em games	Electro-mechanical games - routes via genre:Electromechanical (catlist.ini, not history.xml)
medal games	Medal / prize games (machinetype:medal)
home computers / home consoles	Home computers or consoles (machinetype:computer / machinetype:console)
handheld consoles / handheld games	Handheld gaming systems (machinetype:handheld)
calculators	Electronic calculators (machinetype:calculator)
chess games / chess computers	Chess computers and boards (machinetype:chess board)
musical instruments	Electronic musical instruments (machinetype:musical instrument)
tabletop games / tabletop	Tabletop game units (machinetype:tabletop)
jukeboxes / jukebox	Jukebox machines (machinetype:jukebox)
photo booths / photo booth	Photo booth machines (machinetype:photo booth)
redemption games	Redemption / prize games (machinetype:redemption)

dart games / dart game	Electronic dart games (machinetype:dart game)
bingo games	Bingo machines (machinetype:bingo)
kiddie rides	Kiddie ride machines (machinetype:kiddie ride)
bowlers / bowler	Bowling alley machines (machinetype:bowler)
gun games / gun game	Video gun games (machinetype:gun game)
quiz games / quiz machines	Quiz machines - routed via genre:Quiz (catlist.ini, not history.xml)

SOFTLIST SUPPORT STATUS (3-WAY)

Filter software list items by their MAME support level using supportstatus: (alias: ss:). Returns no results for arcade machines - combine with explicit system scope for best results, e.g. system:snes supportstatus:yes.

supportstatus:yes / ss:yes	Fully supported softlist items (also: supported games, fully supported)
supportstatus:partial / ss:partial	Partially supported items (also: partially supported, partial support)
supportstatus:no / ss:no	Unsupported items (also: unsupported games, not supported)

COMPOUND NO-X AND RUN-TOGETHER FORMS

Most two-word “no X” and “non-X” phrases also work without the space or hyphen. These are fully equivalent to their spaced forms.

noclones	Same as no clones
nobootleg / nobootlegs	Same as no bootlegs
nomechanical / nonmechanical	Same as not mechanical

noprotection	Same as no protection
noworking	Same as nonworking
noarcade	Same as non-arcade (IsArcade = false)
nosamples	Same as no samples
nolightgun	Same as no lightgun
nodevices	Same as no devices
nosoftlist / nosoftware / nomedia	Same as machines only
nobios / nonbios	Same as no bios
coinop / coin op	Same as arcade only
coop	Same as co-op (multiplayer)
vertical only / horizontal only	Same as vertical / horizontal
noX (general pattern)	Most 'no X' and 'non-X' phrases support a compact no-space form. If 'no clones' works, 'noclones' works too.

DATE & DECADE FILTERING

Natural language date expressions are parsed automatically.

1982	Games from 1982, or with 1982 in the title
79-82 / 1979-1983	Year range (2-digit or 4-digit, or mixed: 1999-02)

after 1990	Games released after 1990
before 1985 / pre-1985	Games released before 1985
made in 1982	Games from exactly 1982 (natural sentence form)
from 1980 to 1985	Inclusive year range
between 1991 and 1995	Inclusive year range (advanced syntax form)
80s games / in the 90s	Full decade (1980–1989, 1990–1999, etc.)
early 90s	First half of decade (1990–1993)
late 80s	Last third of decade (1987–1989)
00s games / in the 2010s	Works for any decade including 2000s and 2010s
mid 80s	Middle of a decade (1984–1986); works for any decade: mid 70s, mid 90s, etc.
since 1990	Open-ended: 1990 and later (Year >= 1990). Also: from 1990
through 1990	Open-ended: 1990 and earlier (Year <= 1990)

FIELD-SPECIFIC SEARCHES

Target a specific data field using field:value syntax. On string fields, colon (:) performs a case-insensitive contains match; explicit equals (=) performs exact equality. This distinction lets manufacturer:sega match manufacturer strings containing Sega while manufacturer = "Sega" requires the whole value to be exactly Sega.

year:1994	Games from 1994
manufacturer:capcom	Capcom games (contains match)

name:dkong	Games whose internal name contains 'dkong'
description:puzzle	Games with 'puzzle' in the title/description
cloneof:pacman	Clones of Pac-Man
sourcefile:cps2.cpp	Games on the CPS-2 hardware driver (aliases: hardware:cps2 or driver:cps2)
media:cartridge	Cartridge software (softlist items) (values: cartridge, cassette, floppy, cdrom, harddisk, memcard, rom)
DriverStatus:imperfect	Games with imperfect driver status
workingstatus:working / :imperfect / :notworking	3-way machine driver status filter (alias: ws:). Machines only. Finer-grained than IsWorking: which treats good and imperfect as equivalent
supportstatus:yes / :partial / :no	3-way softlist support level filter (alias: ss:). Softlist items only. Combine with explicit system scope: system:snes ss:yes
machinetype:fruit machine	Substring match against machine-type descriptor from history.xml (aliases: mtype:, mt:). Examples: mtype:computer, mt:pinball
mameversion:0.285 / mv:0.285	Exact match on MAME first-appearance version from MAMEInfo.dat. Also: version 285 (natural form - integer only)
mameinfo:scramble / mi:scramble	Search full MAMEInfo.dat descriptive text for machines; distinct from mameversion: first-appearance filtering
history:scramble / hist:scramble	Search History.xml descriptive text for the given term. Matches both arcade machines and software list items
genre:Shooter / gg:Fighter	Substring match against CatList.ini category. Catches all subtypes automatically
HasLightgun:true	Direct boolean field access

IsArcade:false	Non-arcade machines
IsWorking:true	Broad working-state field: good/imperfect machines and yes/partial softlist items
IsFavorite:true	Favorite games (direct field form)
UsesSamples:true	Games requiring sample files
CloneOf:""	Parent games (empty CloneOf = not a clone)

DAT prefix fields need their colon: genre:fighting, history:pacman, machinetype:pinball. Without it the words are ordinary text, so gg aleste finds GG Aleste. workingstatus imperfect and supportstatus partial still work without the colon for their listed values. Multi-word values are captured correctly: machinetype:quiz machine finds quiz-type machines without 'machine' leaking into text search.

GENRE ALIASES

Genre keywords translate to genre: queries against catlist.ini (requires CatList.ini configured in Art Paths). Substring matching is used, so genre:Shooter catches all subtypes - Flying Vertical, Flying Horizontal, Run and Gun, etc. These aliases combine freely with other filters: shmups vertical working, brawlers noclones since 1990. A one-word alias such as shmups or golf also matches titles containing the word; the phrase forms (shoot em up, golf games) are the genre alone.

shmups / shoot em up / shooters	Shoot-'em-ups and all shooter subtypes (genre:Shooter via catlist.ini)
fighters / fighting games / vs fighters	All fighting game subtypes (genre:Fighter)
platformers / platform games / run and jump	Platform / jump-and-run games (genre:Platform)
brawlers / beat em ups / belt scrollers	Beat-'em-up / brawler games (genre:Beat)
racing / racing games / driving games	Racing and driving games (genre:Racing)

puzzle / puzzle games / puzzlers	Puzzle games (genre:Puzzle)
maze / maze games	Maze games (genre:Maze)
rpg / role playing	Role-playing games (genre:RPG)
sports / sports games	Sports games (genre:Sports)
adventure / adventure games	Adventure games (genre:Adventure)
casino / casino games	Casino games (genre:Casino)
baseball / basketball / soccer / boxing	Sport-specific catlist subgenres (e.g. genre:Baseball, genre:Boxing)

DIAGNOSTIC & QUALITY FILTERS

These filters target ROM and disk dump quality and diagnostic cross-queries. Bad dump and no dump data are populated from the MAME XML at build time. Prototype detection uses a description substring match.

bad dump / bad dumps / bad roms	Machines with at least one known bad ROM or disk dump (HasBadDumps = true)
no dump / nodump / missing dump	Machines with at least one ROM or disk dump not yet found (HasNoDumps = true)
clean roms / good dumps / verified roms	No bad or missing dumps (HasBadDumps = false AND HasNoDumps = false)
prototype / prototypes / proto	Prototype and pre-release versions (description contains 'proto')
no prototypes / no proto	Exclude all prototype / pre-release entries
working clones of nonworking parents	Working clones whose parent set is not working (diagnostic cross-query)

nonworking parents with working clones	Non-working parents that have at least one working clone
working bios / nonworking bios	BIOS entries filtered by working status
working mechanical clones	Working clone entries that are also mechanical machines
nonworking mechanical parents	Non-working parent entries that are mechanical machines

FIELD NEGATION (!)

Prefix the value with ! to exclude matches.

manufacturer:!capcom	Every manufacturer EXCEPT Capcom
year:!1994	Every year EXCEPT 1994

ADVANCED OPERATORS

For precise control, use the SQL-like operators below, mixed with words, quoted phrases and keyword modifiers. Operands keep the order typed. Words side by side apply to the whole explicit OR group: model2 OR daytona nonworking means (model2 OR daytona) AND nonworking. An explicitly typed AND binds more tightly than OR, so model2 OR daytona AND nonworking restricts only the second alternative. Use parentheses to make the intended grouping explicit. A trailing operator with no operand, such as model2 OR, is ignored. Word operators must be uppercase: AND, OR, NOT, IN, LIKE and BETWEEN, including the AND between BETWEEN bounds. Lowercase and mixed-case words remain prose even beside a field expression or parentheses. IN, LIKE and BETWEEN also require a field name and operands, as in the examples below. Natural-language phrases such as not working, between 1991 and 1995 and game and watch remain valid. Field names and search values remain case-insensitive; symbolic operators are unchanged. Quote a field value containing uppercase operator words.

=	Equality	manufacturer = "Capcom"
!=	Inequality	manufacturer != "Sega"
> >=	Greater than / or equal	year > 1990

< <=	Less than / or equal year <= 1985
LIKE	Substring match (% is wildcard) description LIKE "%Fighter%"
NOT LIKE	Substring exclusion description NOT LIKE "%bootleg%"
BETWEEN	Inclusive numeric range year BETWEEN 1990 AND 1992
IN	Matches any value in a list manufacturer IN ("Taito", "Irem")
AND	Both conditions must be true working AND year > 1990
OR	Either condition can be true manufacturer = "Sega" OR manufacturer = "Namco"
NOT	Negates a condition NOT (manufacturer = "Capcom")
( )	Groups expressions (manufacturer = "Sega" OR manufacturer = "Namco")

PUTTING IT ALL TOGETHER

Any combination of the above modes works in a single query. Keywords are translated to advanced syntax automatically before the filter runs.

Working, parent-only Midway games:

```
midway working only parents only
```

Capcom games from the late 90s, hiding Street Fighter:

```
capcom late 90s description NOT LIKE "%Street Fighter%"
```

All Sega or Namco games after 1990:

year > 1990 AND (manufacturer = "Sega" OR manufacturer = "Namco")

Parent puzzle games from 1995 or later:

Year >= "1995" noclones description LIKE "%puzzle%"

NES cartridge games (software list):

nes cartridges

Working 6-button arcade games, no clones, after 1991:

6 buttons working only noclones year > 1991 arcade only

Diagnostic: find playable versions of broken sets:

working clones of nonworking parents

All imperfect Sega games from the 90s:

sega workingstatus:imperfect 90s

Supported SNES cartridges only:

snes cartridges supportstatus:yes

Fruit machines added in MAME 0.196 or later:

fruit machines mameversion >= 0.196

USER GUIDE VERSION HISTORY

Date	Version	Document
Sunday, December 17, 2006	Rev. a	Concept Doc
Tuesday, June 26, 2007	Rev. b	Design Doc
Saturday, April 09, 2011	Rev. c	Design Doc
Sunday, May 26, 2011	1.0	User Guide
Sunday, June 12, 2011	1.5	User Guide
Sunday, October 16, 2011	1.5.3	User Guide

Wednesday, November 02, 2011	1.5.5	User Guide
Sunday, November 20, 2016	1.7	User Guide
Tuesday, January 15, 2019	1.8.1.2	User Guide
Saturday, April 17, 2021	1.8.4	User Guide
Friday, May 5, 2023	1.8.5	User Guide
Thursday, September 30, 2025	2.1.1	User Guide
Wednesday, October 15, 2025	2.2.2	User Guide
Thursday, November 27, 2025	2.4.7	User Guide
Saturday, December 6, 2025	2.4.9	User Guide
Friday, December 26, 2025	2.4.10	User Guide
Monday, February 23, 2026	2.4.14	User Guide
Wednesday, March 25, 2026	2.6.2	User Guide
Sunday, May 24, 2026	2.8.4	User Guide
Saturday, June 13, 2026	2.8.8	User Guide
Sunday, August 2, 2026	2.8.10	User Guide
Friday, October 2, 2026	2.8.12	User Guide
Saturday, October 10, 2026	2.8.14	User Guide
Sunday, November 8, 2026	2.16	User Guide
Monday, November 9, 2026	2.17	User Guide

CREDITS

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